

Reviewer Pack — Quantum Entanglement as Tensor Product Rank vs BP_ReadTwice ...

Entry 021c233a68be · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Quantum Entanglement as Tensor Product Rank vs BP_ReadTwice Circuit Threshold

Entry ID: 021c233a68be **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-23 09:28:49 UTC

1. Conjecture

Field A (mathematical branch): Quantum Information Theory (Tensor Network theory) **Field B** (complexity object): Complexity Theory: BP_ReadTwice Width

Statement:

[‘For all instances of a BP with read-twice complexity, the entanglement entropy of the corresponding tensor network is $O(\log \text{size}(P))$, where P is the BP.’, ‘If there exists an instance of a BP with read-twice complexity such that its entanglement entropy is $\Omega(n)$, then the BP_ReadTwice Width of that BP is at least n^2 .’]

Rationale (proposer’s reasoning):

[‘Tensor network theory, particularly in quantum information, provides a way to represent BPs. Entanglement entropy can capture the non-local correlations within a system.’, ‘The conjecture aims to expose a connection between entanglement and circuit complexity, potentially leading to new insights into BP hardness.’]

Taxonomy category: BP_READTWICE (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 8791d914e31e5c72

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for at least 95% of tensor networks corresponding to BPs with read-twice complexity, the entanglement entropy is $O(\log n)$ with an absolute error of $\pm 10\%$. The conjecture is falsified if any seed produces a tensor network with entanglement entropy $\geq n$ or BP_ReadTwice Width $< n^2$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 9 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - ('tensor network theory' AND 'BP_ReadTwice width') - entanglement entropy $0(\log \text{size}(P))$ AND read-twice complexity - $\Omega(n)$ entanglement entropy AND BP_ReadTwice Width $\geq n^2$

Top relevant hits considered: - [http://arxiv.org/abs/2603.01172v1] Midterm Status Report of the ILC Technology Network Activities - [http://arxiv.org/abs/2507.16475v1] On two-dimensional tensor network group symmetries - [http://arxiv.org/abs/1701.04778v2] Tensor network state correspondence and holography - [http://arxiv.org/abs/2605.07582v1] CP violation analysis of local and nonlocal amplitudes in the $\bar{B}^0 \rightarrow \bar{K}^{*0} \mu^+ \mu^-$ decay - [http://arxiv.org/abs/2508.18081v2] GWTC-4.0: Methods for Identifying and Characterizing Gravitational-wave Transients - [http://arxiv.org/abs/2505.00274v1] Future Circular Collider Feasibility Study Report: Volume 2, Accelerators, Technical Infrastructure and Safety - [http://arxiv.org/abs/2510.17487v1] Directional Search for Persistent Gravitational Waves: Results from the First Part of LIGO-Virgo-KAGRA's Fourth Observin - [http://arxiv.org/abs/1411.4413v2] Observation of the rare $B_s^0 \rightarrow \mu^+ \mu^-$ decay from the combined analysis of CMS and LHCb data

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, $\leq 90s$ wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    n = random.randint(5, 40)
    read_twice_complexity = random.randint(1, n-1)

    # Simulate BP and calculate entanglement entropy (simplified model)
    entanglement_entropy = n * math.log(n, 2) + read_twice_complexity

    if entanglement_entropy >= n:
        return {
            "metric_name": "Entanglement Entropy",
            "metric_value": entanglement_entropy,
            "instances_tested": 1,
            "conjecture_holds": False,
            "counterexample": f"BP of size {n} with read-twice complexity {read_twice_complexity}",
        }
```

```

    return {
        "metric_name": "Entanglement Entropy",
        "metric_value": entanglement_entropy,
        "instances_tested": 1,
        "conjecture_holds": True,
        "counterexample": ""
    }

if __name__ == "__main__":
    seeds = [int(x) for x in sys.argv[1:]] if len(sys.argv) > 1 else [2, 3, 5, 7, 11, 13, 17, 19,

    results = []

    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    metric_values = [r["metric_value"] for r in results]
    conjecture_holds_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    mean_metric_value = sum(metric_values) / len(metric_values)
    std_metric_value = math.sqrt(sum((x - mean_metric_value) ** 2 for x in metric_values) / len(me

    if conjecture_holds_fraction >= 0.95:
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_metric_value} support_fraction
    elif any(not r["conjecture_holds"] for r in results):
        first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"{r['counterexample']}\" first_failing_seed={fi
    else:
        print("RESULT: INCONCLUSIVE")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

96255841368'}
TRIAL: {'metric_name': 'Entanglement Entropy', 'metric_value': 147.38196255841368, 'instances_tested
twice complexity 19, entanglement entropy 147.38196255841368'}
TRIAL: {'metric_name': 'Entanglement Entropy', 'metric_value': 108.23866587835397, 'instances_tested
twice complexity 16, entanglement entropy 108.23866587835397'}
TRIAL: {'metric_name': 'Entanglement Entropy', 'metric_value': 91.05865002596161, 'instances_tested
twice complexity 16, entanglement entropy 91.05865002596161'}
TRIAL: {'metric_name': 'Entanglement Entropy', 'metric_value': 61.60335893412778, 'instances_tested
twice complexity 3, entanglement entropy 61.60335893412778'}
TRIAL: {'metric_name': 'Entanglement Entropy', 'metric_value': 20.651484454403228, 'instances_tested
twice complexity 1, entanglement entropy 20.651484454403228'}
TRIAL: {'metric_name': 'Entanglement Entropy', 'metric_value': 128.0964047443681, 'instances_tested
twice complexity 12, entanglement entropy 128.0964047443681'}
TRIAL: {'metric_name': 'Entanglement Entropy', 'metric_value': 115.0419249893113, 'instances_tested
twice complexity 11, entanglement entropy 115.0419249893113'}
TRIAL: {'metric_name': 'Entanglement Entropy', 'metric_value': 30.529325012980813, 'instances_tested
twice complexity 2, entanglement entropy 30.529325012980813'}
TRIAL: {'metric_name': 'Entanglement Entropy', 'metric_value': 210.42124551085624, 'ins

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test crashed before producing data, which prevents us from verifying if the conjecture holds for at least 95% of tensor networks corresponding to BPs with read-twice complexity.
| next: Re-run the test without errors to verify the conjecture’s support condition.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	13577
2	preregistration	ollama_remote	glm4:latest	0	0	9552
3	novelty	ollama_remote	glm4:latest	0	0	8487
4	novelty	ollama_remote	glm4:latest	0	0	9696
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16244
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10176
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7212
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	6872
9	judge	ollama_remote	glm4:latest	0	0	11684

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 93500 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/021c233a68be.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/021c233a68be.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/021c233a68be.tar.gz` (if generated)